

Theoria classica (et quantica?) camporum

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Introductio

Hoc opus scriptum est ut relatio tirocinii initio de theoria classica camporum agentis. Hoc tirocinium magna ex parte in libro *Quantum Field Theory* a Doctore David Tong scripto fundatum est, ergo non mirum est structuram similem passim invenire, vel etiam paraphrasim operis eius (quam plagiarum vocare possimus...). Tamen, cum finis tirocinii nostri esset theoriam classicam camporum discere et ei assuescere, hic invenies fere omnes demonstrationes et definitiones cuiusque obiecti praesentati, exceptis obiectis plerumque communibus et eis quorum demonstratio vel rigorosa definitio vires nostras superabant.

Conventiones

Hic enumeratur summa conventionum quibus magis minusve implicite in reliquo opere utimur. Hae conventiones in tabula praesentabuntur, causa electionis data ubi haec utilis esse potest.

conventio	causa
$\eta^{\mu\nu} = \begin{pmatrix} + & - & - & - \end{pmatrix}$	Haec est conventio magis usitata in contrarium ad $\begin{pmatrix} - & + & + & + \end{pmatrix}$

Tabula 1: Summa conventionum usitarum

1 Theoria lagrangiana camporum

1.1 Aequatio Euleri-Lagrange

Definitio:

Definimus novum genus lagrangiani $\mathcal{L}(\phi, \partial_\mu \phi)$ in campis, quod vocamus: densitatem lagrangianam, per:

$$S = \int L dt = \int \int \mathcal{L} dx^3 dt = \int \mathcal{L} dx^4 \quad (1.1.1)$$

Per abusionem linguae simpliciter lagrangianum vocatur, loco *densitatis* lagrangianae.

Aequatio Euleri-Lagrange:

Sit ϕ_a campus, indice a notatus, talis ut $\delta\phi(\vec{x}, t_1) = \delta\phi(\vec{x}, t_2) = 0$, tunc eius lagrangianum hanc aequationem respectat:

$$\forall a, \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) - \frac{\partial \mathcal{L}}{\partial \phi_a} = 0 \quad (1.1.2)$$

Demonstratio:

Per principium actionis minimae, quaerimus $\delta S = 0$, atqui:

$$\begin{aligned} \delta \mathcal{L} &= \frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta(\partial_\mu \phi_a) \\ &= \frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a \right) - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) \delta \phi_a \\ &= \left(\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) \right) \delta \phi_a + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a \right) \end{aligned} \quad (1.1.3)$$

Ergo, variatio actionis scribitur:

$$\delta S = \int dx^4 \left(\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) \right) \delta \phi_a + \int dx^4 \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a \right) \quad (1.1.4)$$

Quoniam ϕ_a respectat factum quod $\delta\phi_a(\vec{x}, t_1) = \delta\phi_a(\vec{x}, t_2) = 0$, et quod $\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a \right)$ est derivatio totalis, tunc eius integrale nullum est, ergo:

$$\delta S = \int dx^4 \left(\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) \right) \delta \phi_a \quad (1.1.5)$$

et per principium actionis minimae $\delta S = 0$, ex quo sequitur:

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) = 0 \quad (1.1.6)$$

Q.E.D.

1.1.1 Exemplum I: Aequatio Klein-Gordon

Ponantur in spatio Minkowskiano, cuius metrica est $\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(+, -, -, -)$, et consideretur hoc lagrangianum:

$$\mathcal{L} = \frac{1}{2} \eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2 \quad (1.1.1.1)$$

Tunc, habemus:

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) &= \partial_\mu \left(\frac{1}{2} \eta^{\sigma\nu} \left(\underbrace{\frac{\partial (\partial_\sigma \phi)}{\partial (\partial_\mu \phi)}}_{=\delta_\sigma^\mu} \partial_\nu \phi + \underbrace{\frac{\partial (\partial_\nu \phi)}{\partial (\partial_\mu \phi)}}_{=\delta_\nu^\mu} \partial_\sigma \phi \right) \right) \\ &= \partial_\mu \left(\frac{1}{2} \left(\underbrace{\eta^{\sigma\nu} \delta_\sigma^\mu}_{=\eta^{\mu\nu}} \partial_\nu \phi + \underbrace{\eta^{\sigma\nu} \delta_\nu^\mu}_{=\eta^{\sigma\mu}=\eta^{\mu\sigma}} \partial_\sigma \phi \right) \right) \\ &= \partial_\mu \left(\frac{1}{2} \times 2 \underbrace{\eta^{\mu\nu} \partial_\nu \phi}_{=\partial^\mu \phi} \right) \\ &= \partial_\mu \partial^\mu \phi \end{aligned} \quad (1.1.1.2)$$

Et

$$\frac{\partial \mathcal{L}}{\partial \phi} = -m^2 \phi \quad (1.1.1.3)$$

Ergo secundum aequatio (1.1.2):

$$\partial_\mu \partial^\mu \phi + m^2 \phi = 0 \quad (1.1.1.4)$$

1.1.2 Exemplum II: Aequationes Maxwell

In hoc exemplo proponitur aequationes Maxwell in vacuo recuperare.

Ad hoc, consideretur sequens lagrangianum:

$$\mathcal{L} = -\frac{1}{2} (\partial_\mu A_\nu) (\partial^\mu A^\nu) + \frac{1}{2} (\partial_\mu A^\mu)^2 \quad (1.1.2.1)$$

Tunc:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} &= -\frac{1}{2} \left(\underbrace{\frac{\partial (\partial_\rho A_\sigma)}{\partial (\partial_\mu A_\nu)}}_{=\delta_\rho^\mu \delta_\sigma^\nu} \partial^\rho A^\sigma + \eta^{\rho\alpha} \eta^{\sigma\beta} \underbrace{\frac{\partial (\partial_\alpha A_\beta)}{\partial (\partial_\mu A_\nu)}}_{=\delta_\alpha^\mu \delta_\beta^\nu} \partial_\rho A_\sigma \right) + \frac{1}{2} (\eta^{\rho\sigma})^2 \underbrace{\frac{\partial (\partial_\rho A_\sigma)}{\partial (\partial_\mu A_\nu)}}_{=2\delta_\rho^\mu \delta_\sigma^\nu \partial_\rho A_\sigma} \\ &= -\partial^\mu A^\nu + (\partial_\rho A^\rho) \eta^{\mu\nu} \end{aligned} \quad (1.1.2.2)$$

Ergo

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) &= \partial_\mu (-\partial^\mu A^\nu + (\partial_\rho A^\rho) \eta^{\mu\nu}) \\ &= -\partial^2 A^\nu + \partial^\nu \partial_\mu A^\mu \end{aligned} \quad (1.1.2.3)$$

$$\begin{aligned}
&= -\partial_\mu \left(\underbrace{\partial^\mu A^\nu - \partial^\nu A^\mu}_{=F^{\mu\nu}} \right) \\
&= -\partial_\mu F^{\mu\nu}
\end{aligned} \tag{1.1.2.3}$$

Et

$$\frac{\partial \mathcal{L}}{\partial A_\mu} = 0 \tag{1.1.2.4}$$

Ergo secundum aequationes Euleri-Lagrange, invenimus:

$$\partial_\mu F^{\mu\nu} = 0 \tag{1.1.2.5}$$

Demonstrabimus hanc aequationem permittere aequationes Maxwell recuperare (@). Lagrangianum etiam in forma compactiore rescribi potest auxilio tensoris $F^{\mu\nu}$.

$$\begin{aligned}
\mathcal{L} &= -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \\
&= -\frac{1}{4} (\partial_\mu A_\nu - \partial_\nu A_\mu) (\partial^\mu A^\nu - \partial^\nu A^\mu) \\
&= -\frac{1}{4} (\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu - \partial_\nu A_\mu \partial^\mu A^\nu + \partial_\nu A_\mu \partial^\nu A^\mu) \\
&= -\frac{1}{4} (2\partial_\mu A_\nu \partial^\mu A^\nu - 2\partial_\mu A_\nu \partial^\nu A^\mu) \\
&= -\frac{1}{2} (\partial_\mu A_\nu \partial^\mu A^\nu - \eta^{\mu\rho} \partial_\mu A_\nu \partial^\nu A_\rho) \\
&= -\frac{1}{2} (\partial_\mu A_\nu \partial^\mu A^\nu - \partial^\rho A_\nu \partial^\nu A_\rho) \\
&= -\frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu + \frac{1}{2} (\partial^\nu A_\nu)^2
\end{aligned} \tag{1.1.2.6}$$

Q.E.D.

1.2 Invariabilitas Lorentziana

Una ex primis causis cur theoria camporum excogitata sit, fuit reconciliatio mechanicae quanticae cum theoria relativitatis. Ergo, theoriam construere quaerimus quae invariabilis maneat sub transformationibus Lorentzianis:

$$x^\mu \longrightarrow (x^\mu)' = \Lambda^\mu{}_\nu x^\nu \tag{1.2.1}$$

ubi $\Lambda^\mu{}_\nu$ satisfacit:

$$\eta^{\mu\nu} = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \eta^{\rho\sigma} \tag{1.2.2}$$

Hic Λ est matrix transformationum inter duo systemata referentiae,

$$\Lambda^\nu{}_\mu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Rotatio per θ , circa x^3

$$\Lambda^\nu{}_\mu = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Impulsus Lorentzianus secundum x^1
ubi $\gamma = \sqrt{1 - v^2}$

Itaque, volumus campum scalarem sub his transformationibus invariabilem esse, id est, si $\phi(x)$ est solutio aequationum motus, tunc et $\phi'(x) = \phi(\Lambda^{-1}x)$ sit.

Notandum est indicem exponentis -1 adhiberi quia transformationem activam consideramus.

1.2.1 Exemplum I: Aequatio Klein-Gordon

Videamus quomodo lagrangianum primi nostri exempli se habeat sub transformatione:

$$\phi(x) \longrightarrow \phi'(x) = \phi(\Lambda^{-1}x) \quad (1.2.1.1)$$

tunc:

$$\partial_\mu \phi(x) \longrightarrow (\Lambda^{-1})^\nu{}_\mu \partial_\nu \phi(y) \quad (1.2.1.2)$$

ubi $y = \Lambda^{-1}x$, ita lagrangianum sub hac transformatione fit:

$$\begin{aligned} \mathcal{L}'(x) &= \partial_\mu \phi'(x) \partial_\nu \phi'(x) \eta^{\mu\nu} \\ &= (\Lambda^{-1})^\sigma{}_\mu \partial_\sigma \phi(y) \times (\Lambda^{-1})^\rho{}_\nu \partial_\rho \phi(y) \eta^{\mu\nu} \\ &= \partial_\sigma \phi(y) \partial_\rho \phi(y) \underbrace{(\Lambda^{-1})^\sigma{}_\mu (\Lambda^{-1})^\rho{}_\nu \eta^{\mu\nu}}_{=\eta^{\sigma\rho}} \\ &= \partial_\sigma \phi(y) \partial_\rho \phi(y) \eta^{\sigma\rho} = \mathcal{L}(y) \end{aligned} \quad (1.2.1.3)$$

Consequitur ergo actionem invariabilem esse sub hac transformatione:

$$S' = \int dy^4 \mathcal{L}'(y) = \int dy^4 \mathcal{L}(x) = \int dx^4 \mathcal{L}(x) = S \quad (1.2.1.4)$$

Notari potest nos terminum Jacobianum in mutatione $dy \rightarrow dx$ omisisse, quia hic habemus $\det(\Lambda) = 1$.

1.3 Symmetria

1.3.1 Theorema Noether

Theorema Noether:

Si lagrangianum habet symmetriam continuam ex parametro α dependentem, tunc existit flumen (currens) $J_\mu = \sum_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \frac{\delta \phi_a}{\delta \alpha} - F^\mu$, flumen Noetherianum vocatum, symmetriae associatum, quod conservatur, i.e. $\partial_\mu J_\mu = 0$

Demonstratio:

Cum lagrangianum invariabile est sub quadam variatione speciei $\phi \rightarrow \phi + \delta \phi$, tunc haec transformatio vocatur *symmetria*.

Tunc lagrangianum nostrum rescribi potest:

$$\begin{aligned} \delta \mathcal{L} &= \sum_a \left\{ \frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta(\partial_\mu \phi_a) \right\} \\ &= \sum_a \left\{ \frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a \right) - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) \delta \phi_a \right\} \\ &= \sum_a \left\{ \left[\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) \right] \delta \phi_a + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a \right) \right\} \end{aligned} \quad (1.3.1.1)$$

Cum autem aequationes motus satisfiunt, terminus inter uncus evanescit, et ita relinquitur:

$$\delta \mathcal{L} = \sum_a \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a \right) \quad (1.3.1.2)$$

In casu autem symmetriae, variatio lagrangiani secundum parametrum est derivatio totalis quantitatis scalaris:

$$\frac{\delta \mathcal{L}}{\delta \alpha} = \partial_\mu F^\mu \quad (1.3.1.3)$$

Ergo:

$$\partial_\mu \left(\sum_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \frac{\delta \phi_a}{\delta \alpha} - F^\mu \right) \quad (1.3.1.4)$$

Ergo, flumen definire possumus per:

$$J^\mu = \sum_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \frac{\delta \phi_a}{\delta \alpha} - F^\mu \quad (1.3.1.5)$$

Sic relinquitur:

$$\partial_\mu J^\mu = 0 \quad (1.3.1.6)$$

Patet igitur flumen Noether conservari.

Q.E.D.

1.3.1.a Exemplum I: Klein-Gordon

Si consideretur campus scalaris complexus ϕ , cum lagrangiano sequenti:

$$\mathcal{L} = |\partial^\mu \phi|^2 - m^2 |\phi|^2 \quad (1.3.1.7)$$

Quoniam duo sunt gradus libertatis in campo complexo: ϕ et ϕ^* , tunc lagrangianum rescribitur:

$$\mathcal{L} = \partial^\mu \phi \partial_\mu \phi^* - m^2 \phi \phi^* \quad (1.3.1.8)$$

Animadverti potest hoc lagrangianum invariabile esse sub transformatione: $\phi \rightarrow e^{-i\alpha} \phi$, cum $\alpha \in \mathbb{R}$.
Haec transformatio est ergo symmetria lagrangiani, ita habemus:

$$\frac{\delta \phi}{\delta \alpha} = -i\phi \quad \text{et} \quad \frac{\delta \phi^*}{\delta \alpha} = i\phi^* \quad (1.3.1.9)$$

Praeterea lagrangianum immutatum manet, ergo:

$$\frac{\delta \mathcal{L}}{\delta \alpha} = 0 \quad (1.3.1.10)$$

ergo $F^\mu = \text{const}$, et eligere possumus $F^\mu = 0$ sine amissione generalitatis.

Invenitur tunc:

$$\begin{aligned} J^\mu &= -i\phi \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + i\phi^* \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \\ &= -i(\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi) \end{aligned} \quad (1.3.1.11)$$

Et ergo secundum theorema Noether, habemus:

$$\partial_\mu J^\mu = -i(\phi \partial_\mu \partial^\mu \phi^* - \phi^* \partial_\mu \partial^\mu \phi) = 0 \quad (1.3.1.12)$$

Quod etiam per aequationem motus inventam in exemplo I capitis Caput 1.1 verificari potest.

Etenim inveneramus aequationes motus ita scribi:

$$\partial_\mu \partial^\mu \phi = -m^2 \phi \quad (1.3.1.1)$$

Ergo aequatio (1.3.1.11) reducitur ad:

$$J^\mu = im^2(\cancel{\phi \phi^*} - \cancel{\phi^* \phi}) = 0 \quad (1.3.1.2)$$

1.3.1.b Exemplum II: Tensor Energiae-Impulsionis

Si consideretur symmetria translationis per quadrivectorem ξ constantem et valde parvum, scilicet $x^\nu \rightarrow x^\nu + \xi^\nu$, tunc campus rescribi potest:

$$\phi(x + \xi) = \phi(x) + \xi^\nu \partial_\nu \phi(x) + o(\xi) \quad (1.3.1.3)$$

tunc lex transformationis est:

$$\frac{\delta \phi}{\delta \xi^\nu} = \partial_\nu \phi(x) \quad (1.3.1.4)$$

Idem valet pro lagrangiano, etenim:

$$\mathcal{L}(x + \xi) = \mathcal{L}(x) + \xi^\nu \partial_\nu \mathcal{L}(x) \quad (1.3.1.5)$$

ergo

$$\frac{\delta \mathcal{L}}{\delta \xi^\nu} = \partial_\nu \mathcal{L}(x) \quad (1.3.1.6)$$

Habemus ergo flumen Noetherianum pro singulis $\nu = 0, 1, 2, 3$:

$$(J^\mu)_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L} \quad (1.3.1.7)$$

Definimus tunc novum tensorem, tensorem energiae-impulsionis, per:

$$T^\mu_\nu = (J^\mu)_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L} \quad (1.3.1.8)$$

qui verificat, per theorema Noether, hanc proprietatem:

$$\partial_\mu T^\mu_\nu = 0 \quad (1.3.1.9)$$

Hic tensor duas magnitudines magnas in se continet:

- Energiam:

$$\mathcal{E} = T^0_0 = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \dot{\phi} - \mathcal{L} \quad (1.3.1.10)$$

Agnoscutur expressio energiae in mechanica lagrangiana classica, cum:

$$E = \int dx^3 T^0_0 = \int dx^3 \mathcal{E} \quad (1.3.1.11)$$

- Impulsionem:

$$\mathcal{P}^i = T^i_0 \quad (1.3.1.12)$$

cum hic quoque

$$P^i = \int dx^3 \mathcal{P}^i \quad (1.3.1.13)$$

2 Particulae spinus nullius

2.1 Copulatio cum electromagnetismo

Primum demonstrabimus lagrangianum $\mathcal{L} = \partial_\mu \phi \partial^\mu \phi^* - m^2 \phi \phi^*$ invariabile esse sub grupo $U(1)$ quamdiu haec symmetria globalis est. Deinde, cum ad symmetriam localem transibimus, lagrangianum nostrum non iam invariabile erit, ergo quaeremus id denuo invariabile reddere sub hac symmetria $U(1)$ locali.

2.1.1 Praenotanda de grupo $U(1)$

In casu nostro, $U(1)$ isomorpheus est, ergo coincidit cum grupo $(\mathbb{U}, \cdot)$, circulo complexo unitatis multiplicatione praedito, i.e. $\mathbb{U} = \{|z| = 1, z \in \mathbb{C}\} = \{e^{i\alpha}, \alpha \in \mathbb{R}\}$. Hic ergo demonstrabimus $(\mathbb{U}, \cdot)$, et per consequens $U(1)$, vere grupum esse.

Ad hoc, habemus hanc definitionem grupi:

Sit copia G et $\cdot$ lex compositionis internae, tunc $(G, \cdot)$ grupus est, si et solum si:

1. $\exists e \in G, \forall z \in G, ez = ze = z$
2. $\forall z \in G, \exists u \in G, zu = uz = e$
3. $\forall x, y, z \in G, (xy)z = x(yz)$

Videamus quomodo se res habeat in grupo nostro,

sit $z \in \mathbb{U}$, tunc:

- 1 vere est elementum neutrum, etenim

$$|1| = 1 \text{ ergo } 1 \in \mathbb{U} \quad (2.1.1.1)$$

et

$$1 \cdot z = z = z \cdot 1 \quad (2.1.1.2)$$

• z^* vere est inversum ipsius z , etenim:

$$|z^*| = |z|^* = 1^* = 1 \text{ ergo } z^* \in \mathbb{U} \quad (2.1.1.3)$$

et

$$zz^* = z^*z = |z|^2 = 1 \quad (2.1.1.4)$$

• Denique pro $u, v \in \mathbb{U}$, habetur associativitas:

$$(zu)v = zuv = z(uv) \quad (2.1.1.5)$$

Ergo $\mathbb{U}$ vere grupus est, et consequenter etiam $U(1)$.

2.1.2 Symmetria globalis

Hinc porro operabimur cum $U(1) = \{e^{iq\alpha}, \alpha, q \in \mathbb{R}\}$ sicut definitione pro $U(1)$ globali.

Ostendi potest lagrangianum nostrum invariabile esse per elementa huius grupi, id est procedimus ad hanc transformationem:

$$\forall \alpha \in \mathbb{R}, \phi \rightarrow \phi' = e^{iq\alpha} \phi \quad (2.1.2.1)$$

ergo pro $\alpha \in \mathbb{R}$,

$$\begin{aligned} \mathcal{L}' &= \partial_\mu \phi' \partial^\mu \phi'^* - m \phi' \phi'^* \\ &= e^{iq\alpha} e^{-iq\alpha} \partial_\mu \phi \partial^\mu \phi^* - m e^{iq\alpha} e^{-iq\alpha} \phi \phi^* \\ &= \partial_\mu \phi \partial^\mu \phi^* - m \phi \phi^* = \mathcal{L} \end{aligned} \quad (2.1.2.2)$$

Ergo lagrangianum nostrum vere invariabile est per symmetriam globalem inductam a $U(1)$.

Iam demonstravimus (aequatio (1.3.1.11)) sub hac symmetria, flumen Noetherianum inductum fuisse:

$$J^\mu = -iq(\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi) \quad (2.1.2.3)$$

2.1.3 Symmetria localis

Nunc quaeremus symmetriam per $U(1)$ localem reddere, ad hoc, parametrum nostrum α a positione dependere faciemus, i.e.:

$$\alpha \in \mathbb{R} \longrightarrow \alpha \in \mathbb{C}^{\mathbb{R}^4} \quad (2.1.3.1)$$

aliter dictu, α fit functio positionis.

Ita transformatio fit:

$$\forall x \in \mathbb{R}^4, \phi \longrightarrow \phi' = e^{iq\alpha(x)} \phi \quad (2.1.3.2)$$

Proh dolor, derivatio non iam se habet ut oportet:

$$\begin{aligned} \partial_\mu \phi' &= \partial_\mu (e^{iq\alpha} \phi) = \partial_\mu (e^{iq\alpha}) \phi + e^{iq\alpha} \partial_\mu \phi \\ &= iq(\partial_\mu \alpha) e^{iq\alpha} \phi + e^{iq\alpha} \partial_\mu \phi \end{aligned} \quad (2.1.3.3)$$

Propter hoc, lagrangianum nostrum non iam invariabile est, etenim:

$$\begin{aligned} \mathcal{L}' &= \partial_\mu \phi' \partial^\mu \phi'^* - m \phi' \phi'^* \\ &= (iq(\partial_\mu \alpha) e^{iq\alpha} \phi + e^{iq\alpha} \partial_\mu \phi) (-iq(\partial^\mu \alpha) e^{-iq\alpha} \phi^* + e^{-iq\alpha} \partial^\mu \phi^*) - m \phi \phi^* \\ &= q^2 \partial_\mu \alpha \partial^\mu \alpha \phi \phi^* + \underbrace{\partial_\mu \phi \partial^\mu \phi^* - m \phi \phi^*}_{\mathcal{L}} \\ &= \mathcal{L} + q^2 \partial_\mu \alpha \partial^\mu \alpha \phi \phi^* \end{aligned} \quad (2.1.3.4)$$

Ad symmetriam restituendam, oportebit novam derivationem inducere, vocatam *derivationem covariantem*, quam definimus per:

$$D_\mu = \partial_\mu - iqA_\mu \quad (2.1.3.5)$$

ita ut:

$$D_\mu \phi \longrightarrow (D_\mu \phi)' = e^{iq\alpha} D_\mu \phi \quad (2.1.3.6)$$

Ubi A_μ est potentiale vectoriale electromagnetismi, quod hic photonem codificat. Usitate aequationes Maxwell, quas redemonstrabimus in @, invariables sunt per transformationem mensurae $A_\mu \rightarrow A_\mu - \partial_\mu \chi$ ubi χ functio arbitraria est. In hac compage, invenire possumus relationem inter χ et α .

Proprietas:

$$\chi = -\alpha + C \quad (2.1.3.7)$$

ubi c est constans

Demonstratio:

Hic ponimur in casu transformationum sequentium:

$$\begin{aligned} \phi' &= e^{iq\alpha} \phi D_\mu \phi \rightarrow (D_\mu \phi)' = e^{iq\alpha} D_\mu \phi \\ A_\mu &\rightarrow A'_\mu = A_\mu - \partial_\mu \chi \end{aligned} \quad (2.1.3.8)$$

Idea demonstrationis est scribere $(D_\mu \phi)'$ duobus modis, ut deinde ambas expressiones aequemus, sic habemus ex una parte:

$$\begin{aligned} (D_\mu \phi)' &= e^{iq\alpha} D_\mu \phi \\ &= e^{iq\alpha} (\partial_\mu \phi - iq A_\mu \phi) \end{aligned} \quad (2.1.3.9)$$

Et ex altera parte:

$$\begin{aligned} (D_\mu \phi)' &= D'_\mu \phi' = \partial_\mu \phi' - iq A'_\mu \phi' \\ &= iq(\partial_\mu \alpha) e^{iq\alpha} \phi + e^{iq\alpha} \partial_\mu \phi - iq e^{iq\alpha} A_\mu \phi + iq e^{iq\alpha} \partial_\mu \chi \phi \\ &= e^{iq\alpha} (iq(\partial_\mu \alpha) \phi + \partial_\mu \phi - iq A_\mu \phi + iq \partial_\mu \chi \phi) \end{aligned} \quad (2.1.3.10)$$

Invenimus ergo

$$\cancel{\partial_\mu \phi} - iq \cancel{A_\mu \phi} = iq(\partial_\mu \alpha) \phi + \cancel{\partial_\mu \phi} - iq \cancel{A_\mu \phi} + iq \partial_\mu \chi \phi \quad (2.1.3.11)$$

Ergo

$$\partial_\mu \chi = -\partial_\mu \alpha \quad (2.1.3.12)$$

ita bene recuperamus

$$\chi = -\alpha + c \quad (2.1.3.13)$$

Q.E.D.

2.1.4 Aequationes motus cum symmetria locali

Possumus tunc inspicere quid fiat de nostro lagrangiano cum hac nova derivatione, et pro certo habere id vere invariabile esse per U(1) localem.

Ergo lagrangianum nostrum fit:

$$\begin{aligned} \mathcal{L} &= D_\mu \phi D^\mu \phi^* - m^2 \phi \phi^* \\ &= (\partial_\mu - iq A_\mu) \phi (\partial^\mu - iq A^\mu) \phi^* - m^2 \phi \phi^* \\ &= \partial_\mu \phi \partial^\mu \phi^* - iq \partial_\mu \phi A^\mu \phi^* - iq A_\mu \phi \partial^\mu \phi^* - q^2 A_\mu A^\mu \phi \phi^* - m^2 \phi \phi^* \\ &= \partial_\mu \phi \partial^\mu \phi^* - iq (\phi A_\mu \partial^\mu \phi^* + \phi^* A_\mu \partial^\mu \phi) - \phi \phi^* (q^2 A^\mu A_\mu + m^2) \end{aligned} \quad (2.1.4.1)$$

Facile est, auxilio primae formae lagrangiani, demonstrare illud invariabile esse per grupum U(1) localem.

Denique, poterimus inde deducere 3 aequationes motus pro ϕ , ϕ^* et A_μ

1. Pro ϕ :

Habemus ex una parte:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) = \partial_\mu (\partial^\mu \phi^* - iq \phi^* A^\mu) = \square \phi^* - iq \partial_\mu \phi^* A^\mu \quad (2.1.4.2)$$

Et ex altera parte:

$$\frac{\partial \mathcal{L}}{\partial \phi} = -iqA_\mu \partial^\mu \phi^* - \phi^* (q^2 A^\mu A_\mu + m^2) \quad (2.1.4.3)$$

Ergo secundum aequationes Euleri-Lagrange:

$$\square \phi^* + iq(A_\mu \partial^\mu \phi^* - \partial^\mu \phi^* A_\mu) + \phi^* (q^2 A^\mu A_\mu + m^2) = 0 \quad (2.1.4.4)$$

2. Pro ϕ^* : Habemus ex una parte:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^*)} \right) = \partial_\mu (\partial^\mu \phi - iq\phi A^\mu) = \square \phi - iq\partial_\mu \phi A^\mu \quad (2.1.4.5)$$

Et ex altera parte:

$$\frac{\partial \mathcal{L}}{\partial \phi^*} = -iqA^\mu \partial_\mu \phi - \phi (q^2 A^\mu A_\mu + m^2) \quad (2.1.4.6)$$

Ergo secundum aequationes Euleri-Lagrange:

$$\square \phi + iq(A^\mu \partial_\mu \phi - \partial_\mu \phi A^\mu) + \phi (q^2 A^\mu A_\mu + m^2) = 0 \quad (2.1.4.7)$$

3. Pro A_μ : Habemus ex una parte:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) = 0 \quad (2.1.4.8)$$

Et ex altera parte:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial A_\mu} &= -iq(\phi \partial^\mu \phi^* + \phi^* \partial^\mu \phi) - q^2 \phi \phi^* \frac{\partial (A^\nu A_\nu)}{\partial A_\mu} \\ &= -iq(\phi \partial^\mu \phi^* + \phi^* \partial^\mu \phi) - q^2 \phi \phi^* (A^\mu + A^\mu) \\ &= -iq(\phi \partial^\mu \phi^* + \phi^* \partial^\mu \phi) - 2q^2 \phi \phi^* A^\mu \end{aligned} \quad (2.1.4.9)$$

Ergo secundum aequationes Euleri-Lagrange:

$$iq(\phi \partial^\mu \phi^* + \phi^* \partial^\mu \phi) + 2q^2 \phi \phi^* A^\mu = 0 \quad (2.1.4.10)$$

Tunc possumus eas 3 aequationes summam in sequenti tabula exponere:

$$\square \phi + iq(A^\mu \partial_\mu \phi - \partial_\mu \phi A^\mu) + \phi (q^2 A^\mu A_\mu + m^2) = 0 \quad (2.1.4.11)$$

$$\square \phi^* + iq(A_\mu \partial^\mu \phi^* - \partial^\mu \phi^* A_\mu) + \phi^* (q^2 A^\mu A_\mu + m^2) = 0 \quad (2.1.4.12)$$

$$iq(\phi \partial^\mu \phi^* + \phi^* \partial^\mu \phi) + 2q^2 \phi \phi^* A^\mu = 0 \quad (2.1.4.13)$$

Descriptio 1: Aequationes motus pro campo scalari
cum electromagnetismo copulato

3 Particulae spinus 1/2

In paragrapho praecedenti, theoriam pro campis scalaribus explicavimus, quae nobis permisit particulas spinus 0 describere. In hoc capite proponimus explicationem pro particulis spinus semi-integri. Ad hoc, oportebit novum genus obiectorum inducere: spinores.

3.1 Spinores

Huc usque, solum de campis scalaribus egimus, sed pleraeque particulae plures gradus libertatis internae habent. Quod nos cogit considerare campos qui non trivialiter in grupo Lorentziano transformantur. Itaque ad particulas spinus $\frac{1}{2}$ describendas, debemus repraesentationem grupi Lorentziani explicare et studere.

In genere, campi transformantur per:

$$\phi^a(x) \longrightarrow D[\Lambda]^a_b \phi^b(x) \quad (3.1.1)$$

ubi matrices $D[\Lambda]$ sunt repraesentatio grupi Lorentziani.

Cum D sit repraesentatio, illa est morphismus, id est, has proprietates respectat:

$$D[\Lambda_1 \Lambda_2] = D[\Lambda_1] D[\Lambda_2] \quad (3.1.2)$$

$$D[\Lambda^{-1}] = D[\Lambda]^{-1} \quad (3.1.3)$$

$$D[1] = 1 \quad (3.1.4)$$

Ad hanc repraesentationem inveniendam, inspicere possumus quid eveniat pro transformationibus Lorentzianis infinitesimalibus et studere algebrae Lieanae inde resultanti.

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu \quad (3.1.5)$$

ubi ω infinitesimalis est.

Utendo facto quod $\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \eta^{\rho\sigma} = \eta^{\mu\nu}$, tunc sequitur:

$$\begin{aligned} \eta^{\mu\nu} &= (\delta^\mu{}_\rho + \omega^\mu{}_\rho)(\delta^\nu{}_\sigma + \omega^\nu{}_\sigma) \eta^{\rho\sigma} \\ &= \eta^{\mu\nu} + \omega^{\mu\nu} + \omega^{\nu\mu} + \underbrace{\omega^{\mu\nu} \omega^{\nu\mu}}_{=0} \end{aligned} \quad (3.1.6)$$

Ita habemus ut condicionem quod ω antisymmetricus est:

$$\omega^{\mu\nu} = -\omega^{\nu\mu} \quad (3.1.7)$$

Notari potest matrices 4×4 habere $\frac{4 \times (4-1)}{2} = 6$ componentia, sicut pro transformationibus Lorentzianis quae 3 rotationes et 3 impulsus habent, id est 6 transformationes in toto. Ergo transformationes Lorentzianae per matrices 4×4 describi poterunt.

Inducimus ergo matrices $(\mathcal{M}^{\rho\sigma})^{\mu\nu}$ ubi $\mu, \nu \in \llbracket 0; 3 \rrbracket$ indicem matricis repraesentant, $\rho, \sigma \in \llbracket 0; 3 \rrbracket$ indicem axium circum quos vel secundum quos transformatio fit repraesentant:

$$\forall i \in \llbracket 1, 3 \rrbracket, (\mathcal{M}^{0i})^{\mu\nu} \rightarrow \text{Impulsus Lorentzianus secundum axem } x^i$$

$$\forall i, j \in \llbracket 1, 3 \rrbracket, (\mathcal{M}^{ij})^{\mu\nu} \rightarrow \text{Rotatio in plano } (x^i, x^j)$$

Atqui cum ρ et σ sine restrictione, habemus $4 \times 4 = 16$ componentia, quamobrem imponimus ipsi $\mathcal{M}$ ut sit antisymmetrica in ρ et σ , i.e. $\mathcal{M}^{\rho\sigma} = -\mathcal{M}^{\sigma\rho}$, sic remanent solummodo $\frac{4 \times (4-1)}{2} = 6$ componentia. Ponimus ergo hanc basim:

$$(\mathcal{M}^{\rho\sigma})^{\mu\nu} = \eta^{\rho\mu} \eta^{\sigma\nu} - \eta^{\sigma\mu} \eta^{\rho\nu} \quad (3.1.8)$$

Cave, si hanc expressionem per metricam multiplicamus, ut unum ex indicibus deorsum obtineamus, quod in quibusdam calculis utile esse potest, obtinemus:

$$(\mathcal{M}^{\rho\sigma})^\mu{}_\nu = \eta^{\rho\mu} \delta^\sigma{}_\nu - \eta^{\sigma\mu} \delta^\rho{}_\nu \quad (3.1.9)$$

Hae matrices non iam necessario antisymmetrices sunt!

Nunc ergo scribere possumus ω in hac basi:

$$\omega^\mu{}_\nu = \frac{1}{2} \Omega_{\rho\sigma} (\mathcal{M}^{\rho\sigma})^\mu{}_\nu \quad (3.1.10)$$

ubi coefficientis $\frac{1}{2}$ adest ad compensandum factum quod terminos bis sumimus, propter antisymmetriam matricis $\mathcal{M}$, et $\Omega_{\rho\sigma}$ repraesentat id quod transformatio Lorentziana facit.

Tales matrices $\mathcal{M}^{\rho\sigma}$ generatrices transformationum Lorentzianarum vocantur, verificatque relationem algebrae Lieanae Lorentzianae:

$$[\mathcal{M}^{\rho\sigma}; \mathcal{M}^{\tau\nu}] = \eta^{\sigma\tau} \mathcal{M}^{\rho\nu} - \eta^{\rho\tau} \mathcal{M}^{\sigma\nu} + \eta^{\rho\nu} \mathcal{M}^{\sigma\tau} - \eta^{\sigma\nu} \mathcal{M}^{\rho\tau} \quad (3.1.11)$$

Exemptis indicibus matricum

Demonstratio:

Quaerimus ergo rescribere:

$$[\mathcal{M}^{\rho\sigma}; \mathcal{M}^{\tau\nu}]^\alpha{}_\beta = (\mathcal{M}^{\rho\sigma})^\alpha{}_\beta (\mathcal{M}^{\tau\nu})^\beta{}_\beta - (\mathcal{M}^{\tau\nu})^\alpha{}_\beta (\mathcal{M}^{\rho\sigma})^\beta{}_\beta \quad (3.1.12)$$

Ad hoc, primum unum tantum e duobus terminis calculemus:

$$(\mathcal{M}^{\rho\sigma})^\alpha{}_\beta (\mathcal{M}^{\tau\nu})^\beta{}_\beta = (\eta^{\rho\alpha} \delta^\sigma{}_\beta - \eta^{\sigma\alpha} \delta^\rho{}_\beta) (\eta^{\tau\beta} \delta^\nu{}_\beta - \eta^{\nu\beta} \delta^\tau{}_\beta) \quad (3.1.13)$$

$$\begin{aligned}
&= \eta^{\rho\alpha} \eta^{\tau\sigma} \delta^\nu_\beta - \eta^{\rho\alpha} \eta^{\nu\sigma} \delta^\tau_\beta \\
&\quad + \eta^{\sigma\alpha} \eta^{\nu\rho} \delta^\tau_\beta - \eta^{\sigma\alpha} \eta^{\tau\rho} \delta^\nu_\beta
\end{aligned} \tag{3.1.13}$$

Faciendo eundem calculum pro $(\mathcal{M}^{\tau\nu})^\alpha_\beta (\mathcal{M}^{\rho\sigma})^\beta_\beta$ (quod aequiparatur permutationi $\rho \leftrightarrow \tau$ et $\sigma \leftrightarrow \nu$), obtenemus:

$$\begin{aligned}
(\mathcal{M}^{\tau\nu})^\alpha_\beta (\mathcal{M}^{\rho\sigma})^\beta_\beta &= \eta^{\tau\alpha} \eta^{\rho\nu} \delta^\sigma_\beta - \eta^{\tau\alpha} \eta^{\nu\sigma} \delta^\rho_\beta \\
&\quad + \eta^{\nu\alpha} \eta^{\sigma\tau} \delta^\rho_\beta - \eta^{\nu\alpha} \eta^{\rho\tau} \delta^\sigma_\beta
\end{aligned} \tag{3.1.14}$$

Sic subtrahendo (inter X et XIII annos maximum) duas quantitates, obtenemus:

$$\begin{aligned}
[\mathcal{M}^{\rho\sigma}; \mathcal{M}^{\tau\nu}]^\alpha_\beta &= (\mathcal{M}^{\rho\sigma})^\alpha_\beta (\mathcal{M}^{\tau\nu})^\beta_\beta - (\mathcal{M}^{\tau\nu})^\alpha_\beta (\mathcal{M}^{\rho\sigma})^\beta_\beta \\
&= \underbrace{\eta^{\tau\sigma} (\eta^{\rho\alpha} \delta^\nu_\beta - \eta^{\nu\alpha} \delta^\sigma_\beta)}_{=(\mathcal{M}^{\rho\nu})^\alpha_\beta} - \underbrace{\eta^{\rho\tau} (\eta^{\sigma\alpha} \delta^\nu_\beta - \eta^{\nu\alpha} \delta^\sigma_\beta)}_{=(\mathcal{M}^{\sigma\nu})^\alpha_\beta} \\
&\quad + \underbrace{\eta^{\rho\nu} (\eta^{\sigma\alpha} \delta^\tau_\beta - \eta^{\tau\alpha} \delta^\sigma_\beta)}_{=(\mathcal{M}^{\sigma\tau})^\alpha_\beta} - \underbrace{\eta^{\sigma\nu} (\eta^{\rho\alpha} \delta^\tau_\beta - \eta^{\tau\alpha} \delta^\rho_\beta)}_{=(\mathcal{M}^{\rho\tau})^\alpha_\beta} \\
&= \eta^{\sigma\tau} (\mathcal{M}^{\rho\nu})^\alpha_\beta - \eta^{\rho\tau} (\mathcal{M}^{\sigma\nu})^\alpha_\beta + \eta^{\rho\nu} (\mathcal{M}^{\sigma\tau})^\alpha_\beta - \eta^{\sigma\nu} (\mathcal{M}^{\rho\tau})^\alpha_\beta
\end{aligned} \tag{3.1.15}$$

Q.E.D.

Denique rescribere possumus Λ sub forma exponentiali:

$$\begin{aligned}
\Lambda &= \underbrace{1}_{\substack{=\delta^{\mu\nu} \\ \text{sine indicibus} \\ \text{matricialibus}}} + \omega = 1 + \frac{1}{2} \Omega_{\rho\sigma} \mathcal{M}^{\rho\sigma} \\
&= e^{\frac{1}{2} \Omega_{\rho\sigma} \mathcal{M}^{\rho\sigma}}
\end{aligned} \tag{3.1.16}$$

3.1.1 Algebra Cliffordiana

Ad definiendam repraesentationem spinorum, oportet definire algebram Cliffordianam:

$$\{\gamma^\mu; \gamma^\nu\} = \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu} \mathbf{I}_4 \tag{3.1.1.1}$$

ubi γ^μ matrices sunt et $\mathbf{I}_n$ matrix identitatis dimensionis 4.

aliter dicta γ^μ has proprietates respicit:

- pro $\alpha \neq \beta$

$$\begin{aligned}
\gamma^\alpha \gamma^\beta + \gamma^\beta \gamma^\alpha &= 0 \\
\Rightarrow \gamma^\alpha \gamma^\beta &= -\gamma^\beta \gamma^\alpha
\end{aligned} \tag{3.1.1.2}$$

et

$$\begin{aligned}
2(\gamma^0)^2 &= 2\mathbf{I}_4 \Rightarrow (\gamma^0)^2 = \mathbf{I}_4 \\
\forall i \in \llbracket 1; 3 \rrbracket, 2(\gamma^i)^2 &= -2\mathbf{I}_4 \Rightarrow (\gamma^i)^2 = -\mathbf{I}_4
\end{aligned} \tag{3.1.1.3}$$

Hae matrices necessario dimensionis 4 sunt, etenim in dimensione 2 solum $i\mathbf{I}_2$ valet $-\mathbf{I}_2$ sed adhuc nobis duabus aliis matricibus opus est ad hanc repraesentationem describendam. Similiter in dimensione 3, non difficile est persuadere sibi non satis matricum esse quae $A^2 = -\mathbf{I}_3$ verificant.

In nostro casu, possumus, exempli gratia, has matrices per bloccos considerare:

$$\gamma^0 = \begin{pmatrix} 0_2 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0_2 \end{pmatrix}, \forall i \in \llbracket 1, 3 \rrbracket, \gamma^i = \begin{pmatrix} 0_2 & \sigma^i \\ -\sigma^i & 0_2 \end{pmatrix} \tag{3.1.1.4}$$

ubi σ^i matrices 2×2 sunt, dictae matrices Paulianae, quae sic definiuntur:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{3.1.1.5}$$

Proprietas:

$$\forall i, j \in \llbracket 1; 3 \rrbracket, \{\sigma^i; \sigma^j\} = 2\delta^{ij} \tag{3.1.1.6}$$

Ad comprehendendum quod sit vinculum inter hanc algebram et coetus Lorentzii, demonstrabimus quandam magnitudinem relationem commutationis a aequatio (3.1.14) definitam respicere, ad hoc plures proprietates huius rei demonstrabimus, donec eandem relationem commutationis assequamur. Definiamus igitur $S^{\mu\nu}$ per:

Definitio & proprietates:

$$S^{\rho\sigma} = \frac{1}{4}[\gamma^\rho; \gamma^\sigma] = \begin{cases} 0 & \text{si } \rho = \sigma \\ \frac{1}{2}\gamma^\rho\gamma^\sigma & \text{aliter} \end{cases} = \frac{1}{2}\gamma^\rho\gamma^\sigma - \frac{1}{2}\eta^{\rho\sigma} \quad (3.1.1.7)$$

Demonstratio:

• Pro $\mu = \nu$:

$$\frac{1}{4}[\gamma^\mu; \gamma^\mu] = \frac{1}{4}((\gamma^\mu)^2 - (\gamma^\mu)^2) = 0 = \frac{1}{2}(\gamma^\mu)^2 - \frac{1}{2}\eta^{\mu\mu} \quad (3.1.1.8)$$

• Pro $\rho \neq \sigma$:

$$\frac{1}{4}[\gamma^\rho; \gamma^\sigma] = \frac{1}{4} \left(\gamma^\rho\gamma^\sigma - \underbrace{\gamma^\sigma\gamma^\rho}_{=-\gamma^\rho\gamma^\sigma} \right) = \frac{1}{2}\gamma^\rho\gamma^\sigma = \frac{1}{2}\gamma^\rho\gamma^\sigma - \underbrace{\frac{1}{2}\eta^{\rho\sigma}}_{=0} \quad (3.1.1.9)$$

Q.E.D.

Proprietas:

$$[S^{\mu\nu}, \gamma^\rho] = \gamma^\mu\eta^{\nu\rho} - \gamma^\nu\eta^{\mu\rho} \quad (3.1.1.10)$$

Demonstratio:

$$\begin{aligned} [S^{\mu\nu}, \gamma^\rho] &= \frac{1}{2}((\gamma^\mu\gamma^\nu - \eta^{\mu\nu})\gamma^\rho - \gamma^\rho(\gamma^\mu\gamma^\nu - \eta^{\mu\nu})) \\ &= \frac{1}{2}(\gamma^\mu\gamma^\nu\gamma^\rho - \gamma^\rho\gamma^\mu\gamma^\nu + \cancel{\eta^{\mu\nu}\gamma^\rho} - \cancel{\gamma^\rho\eta^{\mu\nu}}) \\ &= \frac{1}{2} \left(\underbrace{\gamma^\mu(\gamma^\nu\gamma^\rho + \gamma^\rho\gamma^\nu)}_{=\{\gamma^\nu; \gamma^\rho\}} - \underbrace{(\gamma^\mu\gamma^\rho + \gamma^\rho\gamma^\mu)\gamma^\nu}_{=\{\gamma^\mu; \gamma^\rho\}} \right) \\ &= \gamma^\mu\eta^{\nu\rho} - \gamma^\nu\eta^{\mu\rho} \end{aligned} \quad (3.1.1.11)$$

Q.E.D.

Proprietas:

$$[S^{\mu\nu}; S^{\rho\sigma}] = S^{\mu\sigma}\eta^{\nu\rho} - S^{\nu\sigma}\eta^{\mu\rho} + S^{\rho\mu}\eta^{\nu\sigma} - S^{\rho\nu}\eta^{\mu\sigma} \quad (3.1.1.12)$$

Demonstratio:

$$\begin{aligned} [S^{\mu\nu}; S^{\rho\sigma}] &= \frac{1}{2}[S^{\mu\nu}; (\gamma^\rho\gamma^\sigma) - \eta^{\rho\sigma}] \\ &= \frac{1}{2}[S^{\mu\nu}; \gamma^\rho\gamma^\sigma] - \frac{1}{2}\underbrace{[S^{\mu\nu}; \eta^{\rho\sigma}]}_{=0} \\ &= \frac{1}{2}\gamma^\rho[S^{\mu\nu}; \gamma^\sigma] + \frac{1}{2}[S^{\mu\nu}; \gamma^\rho]\gamma^\sigma \\ &= \frac{1}{2}(\gamma^\rho\gamma^\mu\eta^{\nu\sigma} - \gamma^\rho\gamma^\nu\eta^{\mu\sigma} + \gamma^\mu\gamma^\sigma\eta^{\nu\rho} - \gamma^\nu\gamma^\sigma\eta^{\mu\rho}) \end{aligned} \quad (3.1.1.13)$$

Autem secundum aequatio (3.1.1.7): $\gamma^\mu\gamma^\nu = 2S^{\mu\nu} + \eta^{\mu\nu}$, sic:

$$\begin{aligned} [S^{\mu\nu}; S^{\rho\sigma}] &= S^{\rho\mu}\eta^{\nu\sigma} + \cancel{\frac{1}{2}\eta^{\rho\mu}\eta^{\nu\sigma}} - S^{\rho\nu}\eta^{\mu\sigma} - \cancel{\frac{1}{2}\eta^{\rho\nu}\eta^{\mu\sigma}} + S^{\mu\sigma}\eta^{\nu\rho} + \cancel{\frac{1}{2}\eta^{\mu\sigma}\eta^{\nu\rho}} - S^{\nu\sigma}\eta^{\mu\rho} - \cancel{\frac{1}{2}\eta^{\nu\sigma}\eta^{\mu\rho}} \\ &= S^{\rho\mu}\eta^{\nu\sigma} - S^{\rho\nu}\eta^{\mu\sigma} + S^{\mu\sigma}\eta^{\nu\rho} - S^{\nu\sigma}\eta^{\mu\rho} \end{aligned} \quad (3.1.1.14)$$

Q.E.D.

Sic regredimur ad relationem commutationis pro $\mathcal{M}$ pro S , ergo hae duae repraesentationes bene eundem coetum repraesentant, scilicet illum Lorentzii.

3.1.2 Spinores Diraci

Spinores Diraci definiuntur per campum spinorialem, qui res quattuor componentium complexarum est et qui ut sequitur per transformationem Lorentzii transformatur:

$$\forall \alpha \in \llbracket 1; 4 \rrbracket, \psi^\alpha \longrightarrow S[\Lambda]^\alpha_\beta \psi^\beta (\Lambda^{-1}x) \quad (3.1.2.1)$$

ubi

$$\begin{aligned} \Lambda &= \exp\left(\frac{1}{2}\Omega_{\rho\sigma}\mathcal{M}^{\rho\sigma}\right) \\ S[\Lambda] &= \exp\left(\frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma}\right) \end{aligned} \quad (3.1.2.2)$$

hic $S[\Lambda]$ repraesentat Λ in repraesentatione basis S scriptum. Notandum quoque est nos isdem Ω pro utraque repraesentatione uti, quod in tuto collocat nos eandem transformationem Lorentzii facere.

Ad confirmandum Λ et $S[\Lambda]$ revera diversa esse, hoc exemplum consideramus:

Exemplum: Rotatio

Consideremus rotationem in plano $\forall i, j \in \llbracket 1, 3 \rrbracket, (x^i, x^j)$, tum, pro $i \neq j$:

$$\begin{aligned} S^{ij} &= \frac{1}{2}\gamma^i\gamma^j = \frac{1}{2}\begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^j \\ -\sigma^j & 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -\sigma^i\sigma^j & 0 \\ 0 & -\sigma^i\sigma^j \end{pmatrix} \\ &= -\frac{i}{2}\varepsilon^{ijk}\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix} \end{aligned} \quad (3.1.2.3)$$

Praeterea parameter Ω in casu rotationis in plano (x^i, x^j) scribi potest:

$$\Omega_{ij} = -\varepsilon_{ijk}\varphi^k \quad (3.1.2.4)$$

ubi $\vec{\varphi} = (\varphi^1, \varphi^2, \varphi^3)$ et $\vec{\sigma} = (\sigma^1, \sigma^2, \sigma^3)$, ubi unaquaeque pars rotationem φ^i circum axem x^i repraesentat

Tum, matrix rotationis fit:

$$\begin{aligned} S[\Lambda] &= \exp\left(\frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma}\right) \\ &= \exp\left(\frac{1}{4}i\varepsilon^{ijk}\varepsilon_{ijl}\varphi^l\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}\right) \\ &= \exp\left(\frac{1}{2}i\delta_l^k\varphi^l\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}\right) \\ &= \exp\left(\frac{1}{2}i\varphi^k\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}\right) \\ &= \begin{pmatrix} \exp(\frac{i}{2}\varphi^k\sigma^k) & 0 \\ 0 & \exp(\frac{i}{2}\varphi^k\sigma^k) \end{pmatrix} \\ &= \begin{pmatrix} e^{i\frac{\vec{\varphi}\cdot\vec{\sigma}}{2}} & 0 \\ 0 & e^{i\frac{\vec{\varphi}\cdot\vec{\sigma}}{2}} \end{pmatrix} \end{aligned} \quad (3.1.2.5)$$

Sic in casu rotationis θ circum axem x^3 , $\vec{\varphi} = (0, 0, \theta)$, tum:

$$S[\Lambda] = \begin{pmatrix} e^{i\frac{\theta\sigma^3}{2}} & 0 \\ 0 & e^{i\frac{\theta\sigma^3}{2}} \end{pmatrix} \quad (3.1.2.6)$$

sic in casu rotationis θ , spinor solum dimidium huius rotationis efficiet, exempli gratia in casu rotationis integrae ($\theta = 2\pi$), habemus:

$$S[\Lambda] = -I_4 \quad (3.1.2.7)$$

et ergo spinor hanc transformationem sequetur:

$$\psi^\alpha(x) \longrightarrow -\psi^\alpha(x) \quad (3.1.2.8)$$

Ad hoc ut res rotationem integram faciat, oportet igitur eam rotationem 4π facere.

Hae res sunt comparative contra-intuitivae, sed plura exempla accessibilia exstant, sicut rotatio manus sicut in saltatione Tari Piring, aut cum circum taeniam Moebii is (quod nihil ad vaccinia pertinet !), quo in casu cum gyrum perfeceris, in altera parte folii te reperis!

Cave !

Repraesentatio coetus Lorentzii non est unitaria, etenim ut sit, oportet $S^{\rho\sigma}$ anti-hermitianam esse, i.e. $(S^{\rho\sigma})^\dagger = -S^{\rho\sigma}$, sed habemus:

$$(S^{\rho\sigma})^\dagger = \frac{1}{4} [(\gamma^\rho)^\dagger; (\gamma^\sigma)^\dagger] \quad (3.1.2.9)$$

Nisi quod $(\gamma^0)^\dagger = \gamma^0$ et $(\gamma^i)^\dagger = -\gamma^i$, ergo repraesentatio nostra unitaria esse non potest. Revera, hunc eventum generalizare possumus monstrando nullam repraesentationem unitariam dimensionis finitae coetuum Lorentzii exstare.

3.2 Actio Diraci

Nunc propositum est actionem construere quae invariants sit per transformatas Lorentzii, super novis campis quos modo construximus: spinores Diraci. Ad hoc quaeritur actio quae scalaris sit per transformatas Lorentzii.

Simpliciter, spectare poterimus quomodo $\psi^\dagger(x)\psi(x)$ transformetur, ubi $\psi^\dagger$ definitur per:

$$\psi^\dagger(x) = (\psi^*)^t(x) \quad (3.2.1)$$

Spectemus igitur, quomodo hoc productum sub hac transformatione transformetur:

$$\begin{aligned} \psi &\longrightarrow S[\Lambda]\psi(\Lambda^{-1}x) \\ \psi^\dagger &\longrightarrow \psi^\dagger(\Lambda^{-1}x)S[\Lambda]^\dagger \end{aligned} \quad (3.2.2)$$

Tum productum fit:

$$(\psi')^\dagger(x)\psi'(x) = \psi^\dagger(\Lambda^{-1}x)S[\Lambda]^\dagger S[\Lambda]\psi(\Lambda^{-1}x) \quad (3.2.3)$$

Sed monstravimus (disputavimus) repraesentationem coetus Lorentzii non esse unitariam ergo $S[\Lambda]^\dagger S[\Lambda] \neq 1$, sic hoc productum non ut scalaris transformatur, et non habet bonam transformationem in coetu Lorentzii. Nunc cum viderimus cur hoc non procedat. Ad elementum construendum quod procedit, consideremus algebram Cliffordianam, talem ut: $(\gamma^0)^\dagger = \gamma^0$ et $\forall i \in \llbracket 1, 3 \rrbracket$, $(\gamma^i)^\dagger = -\gamma^i$, sic, primum notare poterimus:

$$\begin{aligned} \gamma^0\gamma^0\gamma^0 &= \gamma^0 = (\gamma^0)^\dagger \\ \forall i \in \llbracket 1, 3 \rrbracket, \gamma^0\gamma^i\gamma^0 &= -\gamma^i = (\gamma^i)^\dagger \end{aligned} \quad (3.2.4)$$

Sic, has formulas in unam multo commodiorem rescribere possumus:

$$\forall \mu \in \llbracket 0, 3 \rrbracket, (\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0 \quad (3.2.5)$$

Praeterea, habemus:

Proprietas:

$$(S^{\rho\sigma})^\dagger = -\gamma^0 S^{\rho\sigma} \gamma^0 \quad (3.2.6)$$

Demonstratio:

$$(S^{\rho\sigma})^\dagger = \frac{1}{4} [(\gamma^\sigma)^\dagger; (\gamma^\rho)^\dagger] \quad (3.2.7)$$

$$\begin{aligned}
&= \frac{1}{4} [\gamma^0 \gamma^\sigma \gamma^0; \gamma^0 \gamma^\rho \gamma^0] \\
&= -\frac{1}{4} \gamma^0 [\gamma^\rho; \gamma^\sigma] \gamma^0 \\
&= -\gamma^0 S^{\rho\sigma} \gamma^0
\end{aligned} \tag{3.2.7}$$

Q.E.D.

Tum, videre possumus quomodo $S[\Lambda]$ sub adiuncto (pugione) se gerat:

Proprietas:

$$(S[\Lambda])^\dagger = \gamma^0 S[\Lambda]^{-1} \gamma^0 \tag{3.2.8}$$

Demonstratio:

$$\begin{aligned}
(S[\Lambda])^\dagger &= \exp\left(\frac{1}{2} \Omega_{\rho\sigma} (S^{\rho\sigma})^\dagger\right) \\
&= \exp\left(-\frac{1}{2} \Omega_{\rho\sigma} \gamma^0 S^{\rho\sigma} \gamma^0\right) \\
&= \underbrace{1}_{=\gamma^0 \gamma^0} - \frac{1}{2} \Omega_{\rho\sigma} \gamma^0 S^{\rho\sigma} \gamma^0 \\
&= \gamma^0 \left(1 - \frac{1}{2} \Omega_{\rho\sigma} S^{\rho\sigma}\right) \\
&= \gamma^0 \exp\left(-\frac{1}{2} \Omega_{\rho\sigma} S^{\rho\sigma}\right) \gamma^0 = \gamma^0 S[\Lambda]^{-1} \gamma^0
\end{aligned} \tag{3.2.9}$$

Q.E.D.

Sic, adiunctum Diraci definire possumus per:

$$\bar{\psi} = (\psi)^\dagger \gamma^0 \tag{3.2.10}$$

Videamus igitur quomodo hic per coetum Lorentzii transformetur:

Proprietas:

$$\bar{\psi}(x) \psi(x) \text{ scalaris Lorentzii est} \tag{3.2.11}$$

Demonstratio:

$$\begin{aligned}
\bar{\psi}'(x) \psi'(x) &= (\psi')^\dagger(x) \gamma^0 \psi'(x) \\
&= \psi^\dagger(\Lambda^{-1}x) S[\Lambda]^\dagger \gamma^0 S[\Lambda] \psi(\Lambda^{-1}x) \\
&= \psi^\dagger(\Lambda^{-1}x) \gamma^0 \cancel{S[\Lambda]^{-1} \gamma^0 \gamma^0 S[\Lambda]} \psi(\Lambda^{-1}x) \\
&= \psi^\dagger(\Lambda^{-1}x) \gamma^0 \psi(\Lambda^{-1}x) \\
&= \bar{\psi}(\Lambda^{-1}x) \psi(\Lambda^{-1}x)
\end{aligned} \tag{3.2.12}$$

Hac eadem methodo, duo alia invariantia per transformatas Lorentzii construere possumus, vectores et tensores, etenim:

Proprietas:

$$\begin{aligned}
&\bar{\psi} \gamma^\mu \psi \text{ vector Lorentzii est, i.e.:} \\
\bar{\psi}(x) \gamma^\mu \psi(x) &\longrightarrow \Lambda^\mu{}_\nu \bar{\psi}(\Lambda^{-1}x) \gamma^\nu \psi(\Lambda^{-1}x)
\end{aligned} \tag{3.2.13}$$

Demonstratio: Vide Tong

Proprietas:

$$\bar{\psi}\gamma^\mu\gamma^\nu\psi \text{ tensor Lorentzii est}$$

$$\bar{\psi}(x)\gamma^\mu\gamma^\nu\psi(x) \longrightarrow \Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma\bar{\psi}(\Lambda^{-1}x)\gamma^\rho\gamma^\sigma\psi(\Lambda^{-1}x) \quad (3.2.14)$$

Demonstratio: Vide Tong

Nunc igitur actionem invariantem per coetum Lorentzii construere possumus, gratia III formarum bilinearium quas modo construximus: $\bar{\psi}\psi$, $\bar{\psi}\gamma^\mu\psi$ et $\bar{\psi}\gamma^\mu\gamma^\nu\psi$, revera solum primae duae utiles erunt, tum actionem Diraci definimus per:

$$S = \int d^4x \bar{\psi}(x)(i\gamma^\mu\partial_\mu - m)\psi(x) \quad (3.2.15)$$

3.3 Aequatio Diraci

Ab actione Diraci, supra definita, hoc Lagrangianum deducere possumus:

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi \quad (3.3.1)$$

Sic habemus:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \psi)} \right) = i\partial_\mu \bar{\psi}\gamma^\mu$$

et

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \bar{\psi})} \right) = 0 \quad (3.3.2)$$

Sed

$$\frac{\partial \mathcal{L}}{\partial \psi} = -m\bar{\psi}$$

et

$$\frac{\partial \mathcal{L}}{\partial \bar{\psi}} = i\gamma^\mu\partial_\mu\psi - m\psi \quad (3.3.3)$$

Ergo per aequationes Euler-Lagrange, has duas aequationes invenimus:

$$(i\gamma^\mu\partial_\mu - m)\psi = 0 \quad (3.3.4)$$

$$i\partial_\mu \bar{\psi}\gamma^\mu + m\bar{\psi} = 0 \quad (3.3.5)$$

Hae aequationes interdum quoque vocantur «radices quadratae» solutionis aequationis Klein-Gordon, etenim hae aequationes aequationi Klein-Gordon solutiones sunt:

$$(i\gamma^\nu\partial_\nu - m)(i\gamma^\mu\partial_\mu + m)\psi = -(\gamma^\mu\gamma^\nu\partial_\nu\partial_\mu + m^2)\psi = 0 \quad (3.3.6)$$

Sed ostendere possumus in nostro casu: $\gamma^\mu\gamma^\nu = \eta^{\mu\nu}$, sic invenimus:

$$-(\eta^{\mu\nu}\partial_\mu\partial_\nu + m^2)\psi = -(\partial^\mu\partial_\mu + m^2)\psi = 0 \quad (3.3.7)$$

haec aequatio non iam de γ^μ dependet, sic eodem modo omnibus componentibus ψ applicatur

Notatio Slash

In multis libris aequatio Diraci ut sequitur scripta invenitur:

$$(i\cancel{\partial} - m)\psi = 0 \quad (3.3.8)$$

ubi hanc notationem, comparative commode, definimus:

$$\cancel{A} = \gamma^\mu A_\mu \quad (3.3.9)$$

3.4 Spinors Chirales

In exemplo vidimus $S[\Lambda]$ in casu rotationis scribi ut:

$$S[\Lambda_{\text{rot}}] = \begin{pmatrix} e^{i\frac{\vec{\varphi} \cdot \vec{\sigma}}{2}} & 0 \\ 0 & e^{i\frac{\vec{\varphi} \cdot \vec{\sigma}}{2}} \end{pmatrix} \quad (3.4.1)$$

Ostendi potest in casu Impulsus (Boost) $S[\Lambda]$ scribi posse:

$$S[\Lambda_{\text{boost}}] = \begin{pmatrix} e^{\vec{\chi} \cdot \vec{\sigma}/2} & 0 \\ 0 & e^{-\vec{\chi} \cdot \vec{\sigma}/2} \end{pmatrix} \quad (3.4.2)$$

ubi $\vec{\chi}$ parameter impulsus est ($\forall i \in \llbracket 1; 3 \rrbracket, \Omega_{i0} = \chi_i$)

Notamus has matrices per bloccos diagonales esse, quod significat repraesentationem spinorum Diraci coetus Lorentzii reductibilem esse, i.e. duo (quoniam duo blocchi in diagonali) subcoetus irreductibiles exstare, nostrae repraesentationi supplementares. Sic spinorem Diraci ψ in duas componentes $u_{\pm}$ decomponere possumus, quas sic definimus:

$$\psi = \begin{pmatrix} u_+ \\ u_- \end{pmatrix} \quad (3.4.3)$$

Duae componentes $u_{\pm}$ vocantur spinores Weyl vel spinores chirales, quia uni blocco nostrae transformationis respondent, transformantur ut:

- pro rotationibus:

$$u_{\pm} \longrightarrow e^{i\vec{\varphi} \cdot \vec{\sigma}/2} u_{\pm} \quad (3.4.4)$$

- pro impulsibus:

$$u_{\pm} \longrightarrow e^{\pm i\vec{\chi} \cdot \vec{\sigma}/2} u_{\pm} \quad (3.4.5)$$

Spectare possumus quid aequatio Diraci fiat, ad hoc in antecessum magnitudines calculemus: $\psi^{\dagger}$ & $\gamma^0 \gamma^{\mu}$

Proprietas:

$$\psi^{\dagger} = \left((u_+)^{\dagger} \quad (u_-)^{\dagger} \right) \quad (3.4.6)$$

Demonstratio:

$$\begin{aligned} \psi^{\dagger} &= \begin{pmatrix} u_+ \\ u_- \end{pmatrix}^{\dagger} = \left(\begin{pmatrix} u_+ \\ u_- \end{pmatrix}^{\star} \right)^t \\ &= \left((u_+^{\star})^t \quad (u_-^{\star})^t \right) = \left(u_+^{\dagger} \quad u_-^{\dagger} \right) \end{aligned} \quad (3.4.7)$$

Q.E.D.

Proprietas:

$$\gamma^0 \gamma^{\mu} = \begin{pmatrix} \bar{\sigma}^{\mu} & 0 \\ 0 & \sigma^{\mu} \end{pmatrix} \quad (3.4.8)$$

ubi $\sigma = (1, \sigma^i)$ et $\bar{\sigma} = (1, -\sigma^i)$ redefinimus, per indicem μ denotatum, quod non est confundendum cum σ^i quod semper ad originales σ refertur

Demonstratio:

$$\gamma^0 \gamma^\mu = \begin{cases} \gamma^0 \gamma^0 = I_4 & \text{si } \mu = 0 \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} = \begin{pmatrix} -\sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix} & \text{aliter} \end{cases}$$

$$= \begin{pmatrix} \bar{\sigma}^\mu & 0 \\ 0 & \sigma^\mu \end{pmatrix} \quad (3.4.9)$$

Q.E.D.

Nunc, parati sumus ad rescribendum Lagrangianum Diraci, pro spinoribus Weyl

$$\begin{aligned} \mathcal{L} &= \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi \\ &= i\psi^\dagger \gamma^0 \gamma^\mu \partial_\mu \psi - m\psi^\dagger \gamma^0 \psi \\ &= i \begin{pmatrix} u_+^\dagger & u_-^\dagger \end{pmatrix} \begin{pmatrix} \bar{\sigma}^\mu & 0 \\ 0 & \sigma^\mu \end{pmatrix} \begin{pmatrix} \partial_\mu u_+ \\ \partial_\mu u_- \end{pmatrix} - m \begin{pmatrix} u_+^\dagger & u_-^\dagger \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_+ \\ u_- \end{pmatrix} \\ &= i \begin{pmatrix} u_+^\dagger & u_-^\dagger \end{pmatrix} \begin{pmatrix} \bar{\sigma}^\mu \partial_\mu u_+ \\ \sigma^\mu \partial_\mu u_- \end{pmatrix} - m \begin{pmatrix} u_+^\dagger & u_-^\dagger \end{pmatrix} \begin{pmatrix} u_- \\ u_+ \end{pmatrix} \\ &= iu_+^\dagger \bar{\sigma}^\mu \partial_\mu u_+ + iu_-^\dagger \sigma^\mu \partial_\mu u_- - m(u_+^\dagger u_- + u_-^\dagger u_+) \end{aligned} \quad (3.4.10)$$

Sed habemus si aequationes motus respiciuntur $\mathcal{L} = 0$, ergo:

$$iu_+^\dagger \bar{\sigma}^\mu \partial_\mu u_+ + iu_-^\dagger \sigma^\mu \partial_\mu u_- - m(u_+^\dagger u_- + u_-^\dagger u_+) = 0 \quad (3.4.11)$$

Sic, notamus in casu fermionis massivi, opus esse u_+ vel u_- describere, quia hi duo termini in termino massae copulantur, tamen, in casu fermionis non massivi hic terminus evanescit, sufficit solum u_+ vel u_- ad tale fermion describendum, quod igitur hanc aequationem motus respicere debet:

$$\begin{aligned} i\bar{\sigma}^\mu \partial_\mu u_+ &= 0 \text{ descriptus a } u_+ \\ i\sigma^\mu \partial_\mu u_- &= 0 \text{ descriptus a } u_- \end{aligned} \quad (3.4.12)$$

Demonstratio:

Cum fermion non massivum sit, tum lagrangianum rescribitur:

$$\mathcal{L} = iu_+^\dagger \bar{\sigma}^\mu \partial_\mu u_+ + iu_-^\dagger \sigma^\mu \partial_\mu u_- \quad (3.4.13)$$

Habemus igitur, ex una parte:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu u_+^\dagger)} \right) = 0 \quad (3.4.14)$$

et ex altera parte:

$$\frac{\partial \mathcal{L}}{\partial u_+^\dagger} = i\bar{\sigma}^\mu \partial_\mu u_+ \quad (3.4.15)$$

Ergo secundum aequationes Euler-Lagrange, invenimus:

$$i\bar{\sigma}^\mu \partial_\mu u_+ = 0 \quad (3.4.16)$$

Secundam aequationem invenimus per eundem calculum per parametrum $u_-^\dagger$

Q.E.D.

3.5 Proiectio in alias repraesentationes

In nostra repraesentatione $S[\Lambda]$ nobis per bloccos diagonalis apparet, ideo enim haec repraesentatio etiam repraesentatio chiralis vocatur, quia hanc decompositionem in spinores chirales explicat. Spectemus quid aliis repraesentationibus γ^μ algebrae Cliffordianae fieret:

$$\gamma^\mu \longrightarrow U\gamma^\mu U^{-1} \text{ et } \psi \longrightarrow U\psi \quad (3.5.1)$$

Sic $S[\Lambda]$ nullam causam habet ut diagonalis per bloccos sit, nunc inveniendum est nobis invariants ad spinores chirales definiendos.

Ad hoc hanc rem definire possumus:

Definitio:

$$\gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3 \quad (3.5.2)$$

Proprietas:

$$\{\gamma^5, \gamma^\mu\} = 0 \text{ et } (\gamma^5)^2 = I_4 \quad (3.5.3)$$

Demonstratio:

sit $\mu \in \llbracket 0; 3 \rrbracket$

$$\begin{aligned} \{\gamma^5, \gamma^\mu\} &= \gamma^5\gamma^\mu + \gamma^\mu\gamma^5 \\ &= -i\gamma^0\gamma^1\gamma^2\gamma^3\gamma^\mu - i\gamma^\mu\gamma^0\gamma^1\gamma^2\gamma^3 \\ &= i\gamma^\mu\gamma^0\gamma^1\gamma^2\gamma^3 - i\gamma^5\gamma^0\gamma^1\gamma^2\gamma^3 = 0 \end{aligned} \quad (3.5.4)$$

Et

$$\begin{aligned} (\gamma^5)^2 &= (-i\gamma^0\gamma^1\gamma^2\gamma^3)(-i\gamma^0\gamma^1\gamma^2\gamma^3) \\ &= -\gamma^0\gamma^1\gamma^2\gamma^3\gamma^0\gamma^1\gamma^2\gamma^3 \\ &= -\gamma^0\gamma^1\gamma^2\gamma^0\gamma^1\gamma^2 \\ &= \gamma^0\gamma^1\gamma^0\gamma^1 \\ &= \gamma^0\gamma^0 = I_4 \end{aligned} \quad (3.5.5)$$

Q.E.D.

Proprietas:

$$[S_{\mu\nu}; \gamma^5] = 0 \quad (3.5.6)$$

Demonstratio:

$$\begin{aligned} S_{\mu\nu}\gamma^5 &= \frac{1}{2}(\eta_{\mu\rho}\eta_{\nu\sigma}\gamma^\rho\gamma^\sigma - \eta_{\mu\nu})\gamma^5 = \frac{1}{2}(\eta_{\mu\rho}\eta_{\nu\sigma}\gamma^\rho\gamma^\sigma\gamma^5 - \gamma^5\eta_{\mu\nu}) \\ &= \frac{1}{2}(-\eta_{\mu\rho}\eta_{\nu\sigma}\gamma^\rho\gamma^5\gamma^\sigma - \gamma^5\eta_{\mu\nu}) \\ &= \frac{1}{2}(\eta_{\mu\rho}\eta_{\nu\sigma}\gamma^5\gamma^\rho\gamma^\sigma - \gamma^5\eta_{\mu\nu}) \\ &= \frac{1}{2}\gamma^5(\eta_{\mu\rho}\eta_{\nu\sigma}\gamma^\rho\gamma^\sigma - \eta_{\mu\nu}) \\ &= \gamma^5 S_{\mu\nu} \end{aligned} \quad (3.5.7)$$

Ergo habemus bene:

$$[S_{\mu\nu}; \gamma^5] = S_{\mu\nu}\gamma^5 - \gamma^5 S_{\mu\nu} = 0 \quad (3.5.8)$$

Q.E.D.

Haec ultima proprietas, nobis indicat γ^5 scalarem sub rotationibus et impulsibus esse et quoniam $(\gamma^5)^2 = I_4$, has projectiones definire possumus, quae igitur per coetum Lorentzii invariantes sunt:

$$P_\pm = \frac{1}{2}(I_4 \pm \gamma^5) \quad (3.5.9)$$

Proprietas:

$$P_{\pm} \text{ est projectio et } P_+ P_- = 0 \quad (3.5.10)$$

Demonstratio:

$$\begin{aligned} (P_{\pm})^2 &= \frac{1}{4}(\mathbf{I}_4 \pm \gamma^5)(\mathbf{I}_4 \pm \gamma^5) \\ &= \frac{1}{4} \left(\mathbf{I}_4 \pm 2\gamma^5 + \underbrace{(\gamma^5)^2}_{=\mathbf{I}_4} \right) \\ &= \frac{1}{4}(2\mathbf{I}_4 \pm 2\gamma^5) \\ &= \frac{1}{2}(\mathbf{I}_4 \pm \gamma^5) = P_{\pm} \end{aligned} \quad (3.5.11)$$

Et

$$\begin{aligned} P_+ P_- &= (\mathbf{I}_4 + \gamma^5)(\mathbf{I}_4 - \gamma^5) \\ &= \mathbf{I}_4 - (\gamma^5)^2 = \mathbf{I}_4 - \mathbf{I}_4 = 0 \end{aligned} \quad (3.5.12)$$

Q.E.D.

In nostra repraesentatione γ^5 scribitur:

$$\begin{aligned} \gamma^5 &= -i\gamma^0\gamma^1\gamma^2\gamma^3 \\ &= -i \begin{pmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^1 \\ -\sigma^1 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^2 \\ -\sigma^2 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^3 \\ -\sigma^3 & 0 \end{pmatrix} \\ &= -i \begin{pmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^1 \\ -\sigma^1 & 0 \end{pmatrix} \begin{pmatrix} -\sigma^2\sigma^3 & 0 \\ 0 & -\sigma^2\sigma^3 \end{pmatrix} \\ &= -i \begin{pmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^1 \\ -\sigma^1 & 0 \end{pmatrix} \begin{pmatrix} -i\sigma^1 & 0 \\ 0 & -i\sigma^1 \end{pmatrix} \\ &= - \begin{pmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{pmatrix} \begin{pmatrix} 0 & (\sigma^1)^2 \\ -(\sigma^1)^2 & 0 \end{pmatrix} \\ &= - \begin{pmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{pmatrix} \begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix} \\ &= - \begin{pmatrix} -\mathbf{I}_2 & 0 \\ 0 & \mathbf{I}_2 \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{I}_2 & 0 \\ 0 & -\mathbf{I}_2 \end{pmatrix} \end{aligned} \quad (3.5.13)$$

Notamus igitur in nostra repraesentatione $P_{\pm}$ spinorem Diraci ψ in eius componentes chirales proicere. Componentes chirales igitur pro qualibet repraesentatione definire possumus per:

$$\psi_{\pm} = P_{\pm} \psi \quad (3.5.14)$$

quae igitur repraesentationes irreductibiles coetus Lorentzii sunt.

3.5.1 Paritas

In mechanica quantica, duae symmetriae discretiae comparative graves sunt:

Reversio temporis: $(x^0, x^i) \longrightarrow (-x^0, x^i)$

$$\text{Paritas: } (x^0, x^i) \longrightarrow (x^0, -x^i) \quad (3.5.1.1)$$

Nos tantum de symetria paritatis curabimus. Spectemus quomodo igitur spinores chirales transformantur, ad hoc spectabimus quid fiat pro elementis coetus $(\frac{1}{2}, 0)$, haec elementa igitur definiuntur sicut ita transformantia:

- Rotatio:

$$S[\Lambda_{\text{Rot}}]u = e^{\frac{i}{2}\vec{\varphi} \cdot \vec{\sigma}}u \quad (3.5.1.2)$$

- Impulsus:

$$S[\Lambda_{\text{Boost}}]u = e^{\frac{1}{2}\vec{\chi} \cdot \vec{\sigma}}u \quad (3.5.1.3)$$

Sed cum projectionem his transformationibus applicamus obtinemus:

- Rotatio:

$$P(S[\Lambda_{\text{Rot}}]u) = e^{\frac{i}{2}\vec{\varphi} \cdot \vec{\sigma}}u \quad (3.5.1.4)$$

- Impulsus:

$$P(S[\Lambda_{\text{Boost}}]u) = e^{-\frac{1}{2}\vec{\chi} \cdot \vec{\sigma}}u \quad (3.5.1.5)$$

Quod exacte definitio elementi coetus $(0, \frac{1}{2})$ est, eadem methodo ostendere possemus elementum coetus $(0, \frac{1}{2})$ fieri elementum coetus $(\frac{1}{2}, 0)$, sic P bijectionem inter $(0, \frac{1}{2})$ et $(\frac{1}{2}, 0)$ efficit, aliter dictum, habemus:

$$P(u_{\pm}) = u_{\mp} \quad (3.5.1.6)$$

Sic spinor Diraci transformatur per:

Proprietas:

$$P(\psi) = \gamma^0 \psi \quad (3.5.1.7)$$

Demonstratio:

$$\begin{aligned} \gamma^0 \psi &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = \begin{pmatrix} \psi_- \\ \psi_+ \end{pmatrix} \\ &= \begin{pmatrix} P(\psi_+) \\ P(\psi_-) \end{pmatrix} = P\left(\begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}\right) = P(\psi) \end{aligned} \quad (3.5.1.8)$$

Q.E.D.

Spectare possumus quid fiat aequatio Diraci per hanc symetriad:

$$(i\gamma^\mu \partial_\mu - m)\psi' = (i\gamma^0 \partial_t + i\gamma^i \partial_i - m)\gamma^0 \psi = \gamma^0 (i\gamma^0 \partial_t - \gamma^i \partial_i - m)\psi = 0 \quad (3.5.1.9)$$

ubi signum per transitum γ^5 adeptum compensatur per derivatam super $-\vec{x}$

3.5.1.a Interactivitas choralis

Spectare quoque possumus quomodo termini copulationis sub symetria paritatis transformantur, sic pro quaque forma bilineari ad Caput 3.2 inventa calculabimus.

Pro scalaribus, habemus:

$$P(\bar{\psi}\psi) = \overline{P(\psi)}P(\psi) = \overline{\gamma^0 \psi} \gamma^0 \psi = \bar{\psi} \gamma^0 \gamma^0 \psi = \bar{\psi}\psi \quad (3.5.1.1)$$

Quod transformatio scalarum est.

Pro vectore, casum temporalem a casu spatiali distinguemus:

$$\begin{aligned} P(\bar{\psi}\gamma^0 \psi) &= \overline{P(\psi)} \gamma^0 P(\psi) = \bar{\psi} \gamma^0 \gamma^0 \gamma^0 \psi = \bar{\psi} \gamma^0 \psi \\ P(\bar{\psi}\gamma^i \psi) &= \overline{P(\psi)} \gamma^i P(\psi) = \bar{\psi} \gamma^0 \gamma^i \gamma^0 \psi = -\bar{\psi} \gamma^i \psi \end{aligned} \quad (3.5.1.2)$$

Observamus, ut expectatum, inversionem signi super componentibus spatialibus vectorum, et quod haec bene sicut vectores transformantur.

Ostendere possemus tensorem $S^{\mu\nu}$ quoque sicut decet transformari, tamen, magis curabimus de duabus formis bilinearibus quas in Caput 3.5 ope γ^5 construere potuimus: $\bar{\psi}\gamma^5\psi$ & $\bar{\psi}\gamma^5\gamma^\mu\psi$

$$P(\bar{\psi}\gamma^5\psi) = \overline{P(\psi)}\gamma^5P(\psi) = \bar{\psi}\gamma^0\gamma^5\gamma^0\psi = -\bar{\psi}\gamma^5\psi$$

$$P(\bar{\psi}\gamma^5\gamma^\mu\psi) = \overline{P(\psi)}\gamma^5\gamma^\mu P(\psi) = \bar{\psi}\gamma^0\gamma^5\gamma^\mu\gamma^0\psi = \begin{cases} -\bar{\psi}\gamma^5\gamma^0\psi & \text{si } \mu = 0 \\ +\bar{\psi}\gamma^5\gamma^\mu\psi & \text{si } \mu \neq 0 \end{cases} \quad (3.5.1.1)$$

Habemus igitur $\bar{\psi}\gamma^5\psi$ sicut pseudo-scalarem transformari et $\bar{\psi}\gamma^5\gamma^\mu\psi$ sicut pseudo-vectorem transformari

Typum diversarum formarum bilinearium quas hucusque construximus in tabula sequenti comprehendere possumus:

$\bar{\psi}\psi$	scalaris
$\bar{\psi}\gamma^\mu\psi$	vector
$\bar{\psi}S^{\mu\nu}\psi$	Tensor
$\bar{\psi}\gamma^5\psi$	pseudo-scalaris
$\bar{\psi}\gamma^5\gamma^\mu\psi$	pseudo-vector

Descriptio 2: Typus formarum bilinearium

Invenimus igitur $1 + 4 + (4 \times 3/2) + 1 + 4 = 16$ formas bilineares, qui igitur numerus maximus est quem invenire sperare possumus.

3.6 Symetria & Fluxio conservata

Lagrangianum Diraci 4 symetrias codificat:

1. Translatio spatio-temporalis
2. transformatio Lorentzii
3. symetria interna vectorum
4. symetria axialis

In hoc capite proponimus fluxionem Noetherianam unicuique symetriae associatam calculare. Primum habemus:

$$\frac{\partial \mathcal{L}}{\partial(\partial\psi)} = i\bar{\psi}\gamma^\mu \quad (3.6.1)$$

et

$$\frac{\partial \mathcal{L}}{\partial(\partial\bar{\psi})} = 0 \quad (3.6.2)$$

Ergo habemus:

$$J^\mu = i\bar{\psi}\gamma^\mu \frac{\delta\psi}{\delta\alpha} - F^\mu \quad (3.6.3)$$

3.6.1 Translatio spatio-temporalis

Iam vidimus in casu translationis spatio-temporalis $x \rightarrow x + \varepsilon$, nos habere:

$$\delta\psi = \varepsilon^\nu \partial_\nu \psi \quad (3.6.1.1)$$

Et quod

$$\delta\mathcal{L} = \varepsilon^\nu \partial_\nu \mathcal{L} \quad (3.6.1.2)$$

Sic, fluxio nostra fit:

$$T^\mu{}_\nu = (J^\mu)_\nu = i\bar{\psi}\gamma^\mu\partial_\nu\psi - \delta^\mu{}_\nu\mathcal{L} \quad (3.6.1.3)$$

Sed si aequationes motus respiciuntur, habemus $\mathcal{L} = 0$, unde:

$$T^{\mu\nu} = i\bar{\psi}\gamma^\mu\partial^\nu\psi \quad (3.6.1.4)$$

3.6.2 Transformatio Lorentzii

Sub transformatione Lorentzii infinitesimali, ψ sic transformatur:

$$\begin{aligned} \psi'^\alpha(x) &= S[\Lambda]\psi^\alpha(\Lambda^{-1}x) \\ &= e^{\frac{1}{2}\Omega_{\rho\sigma}(S^{\rho\sigma})^\alpha{}_\beta}\psi^\beta((\delta^\mu{}_\nu - \omega^\mu{}_\nu)x^\nu) \\ &= \left(\delta^\alpha{}_\beta + \frac{1}{2}\Omega_{\rho\sigma}(S^{\rho\sigma})^\alpha{}_\beta\right)\psi^\beta(x^\mu - \omega^\mu{}_\nu x^\nu) \\ &= \left(\delta^\alpha{}_\beta + \frac{1}{2}\Omega_{\rho\sigma}(S^{\rho\sigma})^\alpha{}_\beta\right)(\psi^\beta - \omega^\mu{}_\nu x^\nu\partial_\mu\psi^\beta) \\ &= \psi^\alpha - \omega^\mu{}_\nu x^\nu\partial_\mu\psi^\alpha + \frac{1}{2}\Omega_{\rho\sigma}(S^{\rho\sigma})^\alpha{}_\beta\psi^\beta - \frac{1}{2}\Omega_{\rho\sigma}\omega^\mu{}_\nu(S^{\rho\sigma})^\alpha{}_\beta\psi^\beta \end{aligned} \quad (3.6.2.1)$$

Sed habemus $\omega^\mu{}_\nu = \frac{1}{2}\Omega_{\rho\sigma}(\mathcal{M}^{\rho\sigma})^\mu{}_\nu$ et $(\mathcal{M}^{\rho\sigma})^\mu{}_\nu = \eta^{\rho\mu}\delta^\sigma{}_\nu - \eta^{\sigma\mu}\delta^\rho{}_\nu$, ergo:

$$\begin{aligned} \omega^\mu{}_\nu &= \frac{1}{2}\Omega_{\rho\sigma}(\mathcal{M}^{\rho\sigma})^\mu{}_\nu = \omega^\mu{}_\nu = \frac{1}{2}\Omega_{\rho\sigma}(\eta^{\rho\mu}\delta^\sigma{}_\nu - \eta^{\sigma\mu}\delta^\rho{}_\nu) \\ &= \frac{1}{2}(\Omega^\mu{}_\nu - \Omega_\nu{}^\mu) = \frac{1}{2} \times 2\Omega^\mu{}_\nu = \Omega^\mu{}_\nu \end{aligned} \quad (3.6.2.2)$$

Ergo substituendo $\Omega^{\mu\nu}$ per $\omega^{\mu\nu}$ in aequatio (3.6.2.1) obtenimus:

$$\begin{aligned} \psi'^\alpha(x) &= \psi^\alpha - \omega^\mu{}_\nu x^\nu\partial_\mu\psi^\alpha + \frac{1}{2}\omega_{\rho\sigma}(S^{\rho\sigma})^\alpha{}_\beta\psi^\beta - \underbrace{\frac{1}{2}\omega_{\rho\sigma}\omega^\mu{}_\nu(S^{\rho\sigma})^\alpha{}_\beta\psi^\beta}_{=o(\omega^{\mu\nu})} \\ &= \underbrace{\psi^\alpha - \omega_{\mu\nu}\left(x^\nu\partial^\mu\psi^\alpha - \frac{1}{2}(S^{\mu\nu})^\alpha{}_\beta\psi^\beta\right)}_{=\delta\psi^\alpha} \end{aligned} \quad (3.6.2.3)$$

Ergo

$$\delta\psi^\alpha = -\omega_{\mu\nu}\left(x^\nu\partial^\mu\psi^\alpha - \frac{1}{2}(S^{\mu\nu})^\alpha{}_\beta\psi^\beta\right) \quad (3.6.2.4)$$

Sed per anti-symmetricam primus terminus rescribitur:

$$\begin{aligned} -\omega_{\mu\nu}x^\nu\partial^\mu\psi^\alpha &= -\frac{1}{2}(\omega_{\mu\nu} - \omega_{\nu\mu})x^\nu\partial^\mu\psi^\alpha \\ &= -\frac{1}{2}(\omega_{\mu\nu}x^\nu\partial^\mu\psi^\alpha - \omega_{\nu\mu}x^\mu\partial^\nu\psi^\alpha) \end{aligned} \quad (3.6.2.5)$$

Et ergo habemus tandem:

$$\delta\psi^\alpha = -\frac{1}{2}(\omega_{\mu\nu}x^\nu\partial^\mu\psi^\alpha - \omega_{\nu\mu}x^\mu\partial^\nu\psi^\alpha) + \frac{1}{2}\omega_{\mu\nu}(S^{\mu\nu})^\alpha{}_\beta\psi^\beta \quad (3.6.2.6)$$

Sicut in casu translationis spatio-temporalis habemus $F^\mu = 0$, ergo:

$$\begin{aligned} (J^\mu)^{\rho\sigma} &= i\bar{\psi}\gamma^\mu\left(-\frac{1}{2}(x^\sigma\partial^\rho\psi - x^\rho\partial^\sigma\psi) + \frac{1}{2}(S^{\rho\sigma})\psi\right) \\ &= -\frac{1}{2}x^\sigma\underbrace{i\bar{\psi}\gamma^\mu\partial^\rho\psi}_{=T^{\mu\rho}} + \frac{1}{2}x^\rho\underbrace{i\bar{\psi}\gamma^\mu\partial^\sigma\psi}_{=T^{\mu\sigma}} + \frac{1}{2}\bar{\psi}\gamma^\mu S^{\rho\sigma}\psi \\ &= \frac{1}{2}(x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho} + \bar{\psi}\gamma^\mu S^{\rho\sigma}\psi) \end{aligned} \quad (3.6.2.7)$$

Quoniam fluxiones ad factorem definitae sunt, hanc fluxionem definimus:

$$(\mathcal{J}^\mu)^{\rho\sigma} = 2(J^\mu)^{\rho\sigma} = x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho} + \bar{\psi}\gamma^\mu S^{\rho\sigma}\psi \quad (3.6.2.8)$$

3.6.3 Symetria interna vectorum

Facile est verificare lagrangianum invariants esse per mutationem phasis: $\psi \rightarrow e^{i\alpha}\psi$, sic hoc naturaliter symetriam constituit, sic, habemus:

$$\delta\psi = i\alpha \quad (3.6.3.1)$$

et quoniam lagrangianum invariants est $F^\mu = 0$

Obtinemus igitur hanc fluxionem:

$$\begin{aligned} j^\mu &= i\bar{\psi}\gamma^\mu \times i\psi \\ &= -\bar{\psi}\gamma^\mu\psi \end{aligned} \quad (3.6.3.2)$$

Secundam fluxionem positivam, comparate ad eam a theoremate Noetheriano directe datam, iterum definimus per:

$$j_V^\mu = -j^\mu = \bar{\psi}\gamma^\mu\psi \quad (3.6.3.3)$$

Agnovimus hic expressionem vectoris

3.6.4 Symetria axialis

Cum massa nulla est ($m = 0$) lagrangianum symetriam additam admittit quae fermiones laevos et dextros in sensu opposito rotare facit:

$$\psi \rightarrow e^{i\alpha\gamma^5}\psi \quad (3.6.4.1)$$

Ergo

$$\delta\psi = i\alpha\gamma^5 \quad (3.6.4.2)$$

Sic eodem ratiocinio quo